

Ex2: Bias Audit of Pre-Trained Word Embeddings Using the Word Embedding Association Test (WEAT)

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#EXP 2
import gensim
import numpy as np
from numpy.linalg import norm
import kagglehub

path = kagglehub.dataset_download("leadbest/googlenewsvectornegative300")

print("Path to dataset files:", path)
MODEL_PATH = f"{path}/GoogleNews-vectors-negative300.bin"

print("Loading Google News Word2Vec model...")
model = gensim.models.KeyedVectors.load_word2vec_format(
    MODEL_PATH,
    binary=True
)
print("Model loaded successfully!")
# Target word sets (Gender)
X = ["man", "male", "boy", "father", "he", "him"]
Y = ["woman", "female", "girl", "mother", "she", "her"]

# Attribute word sets
A = ["career", "business", "profession", "office", "salary", "management"]
B = ["family", "home", "children", "parent", "marriage", "house"]

assoc_X = []
assoc_Y = []

# Compute association for X
for w in X:
    sim_A = []
    sim_B = []

    for a in A:
        sim_A.append(
            np.dot(model[w], model[a]) /
            (norm(model[w]) * norm(model[a]))
        )

    for b in B:
        sim_B.append(
            np.dot(model[w], model[b]) /
            (norm(model[w]) * norm(model[b]))
        )
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    assoc_X.append(np.mean(sim_A) - np.mean(sim_B))

# Compute association for Y
for w in Y:
    sim_A = []
    sim_B = []

    for a in A:
        sim_A.append(
            np.dot(model[w], model[a]) /
            (norm(model[w]) * norm(model[a]))
        )

    for b in B:
        sim_B.append(
            np.dot(model[w], model[b]) /
            (norm(model[w]) * norm(model[b]))
        )

    assoc_Y.append(np.mean(sim_A) - np.mean(sim_B))
# WEAT Score
weat_score = np.sum(assoc_X) - np.sum(assoc_Y)

# Effect Size (Cohen's d)
mean_X = np.mean(assoc_X)
mean_Y = np.mean(assoc_Y)
std_dev = np.std(assoc_X + assoc_Y, ddof=1)

effect_size = (mean_X - mean_Y) / std_dev
print("==== WEAT Gender Bias Results =====")
print("WEAT Score:", weat_score)
print("Effect Size:", effect_size)

print("\nInterpretation:")
print("Effect Size > 0 → Male terms more associated with Career")
print("Effect Size < 0 → Female terms more associated with Career")
print("Effect Size ≈ 0 → No detectable bias")
"""

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OUTPUT:

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*** Using Colab cache for faster access to the 'googlenewsvectornegative300' dataset.
Path to dataset files: /kaggle/input/googlenewsvectornegative300
Loading Google News Word2Vec model...
Model loaded successfully!
===== WEAT Gender Bias Results =====
WEAT Score: 0.3700586
Effect Size: 0.5311373

Interpretation:
Effect Size > 0 → Male terms more associated with Career
Effect Size < 0 → Female terms more associated with Career
Effect Size ≈ 0 → No detectable bias
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